InsightDesk - summary of tests for the paper

Support-complaint intelligence: raw complaints are grouped into problem cases, then spikes, cause hypotheses and escalation are computed per case. Every test below had its pass criteria written before it ran. P = precision (share of merges that are correct), R = recall (share of true duplicates found), F1 = their balance; "case P" = how pure each case is.

1. Real data: main results

Test	Data	Result	Why it matters
Real Play Store reviews, human-labelled	338 reviews of 6 help-desk apps, 3 held-out test halves	P **0.81** (95% CI 0.63-0.96), case P ~0.92	Main real-world test; labels by our team (agreement kappa 0.74)
+ case re-joining step (adopted)	same	Recall +26% (0.26 → 0.33), P unchanged	Improvement confirmed on untouched data
Unseen apps (no tuning on them)	588 reviews: Intercom, Jira, ServiceNow	case P **0.92**	Generalises to companies never seen
Unseen company (thresholds from other companies)	Play Store, leave-one-company-out	case P **0.89**	Works for a new customer with no history
One company, product not given	889 Firefox + Thunderbird reports as one stream	P 0.90 vs 0.91 with product; 1 of 157 cases mixed	Companies do not need to supply a product field
IT support tickets (partial run)	Endava tickets, first 350	case P **0.94** vs 0.75 original	Third domain; full 1,000-ticket run pending
Synthetic complaints (sanity check)	488 generated complaints	P 0.98, F1 0.78	Pipeline behaves as designed; never used as headline

2. Second domain: public bug-report benchmarks

Test	Data	Result
Duplicate grouping (thresholds from Play Store only)	Firefox samples 1-3, Thunderbird (Bugzilla duplicate links)	P **0.87-0.96**, R 0.50-0.61; with re-joining Firefox R 0.55 → 0.66, Thunderbird 0.61 → 0.72 (P up to 0.87)
Retrieval on full histories (Recall-Rate@5)	233,521 Firefox, Thunderbird, Eclipse reports (public BugHub split)	0.46-0.56 vs BM25 0.40-0.44; trained retriever up to 0.62

3. Against competitors and published methods (same data, same metric)

Method	Play Store F1	Firefox F1	Note
InsightDesk (+ re-joining)	**0.46**	**0.78**	Decides each complaint on arrival
Original InsightDesk pipeline	0.14 (R1: 0.24, P 0.31 → 0.55, +77%)	-	Significant improvement
CLAP (ICSE 2016), published / tuned	0.05 / 0.24	0.37 / 0.03	Classic app-review grouping
BERTopic topic grouping (used by commercial tools)	0.25	0.54	Topics are not problems (P 0.16)
Embedding threshold only	~0.10	0.07	Typical simple dedup
Fine-tuned small models replacing the LLM	precision 0.13-0.49		The LLM step is necessary

4. Against GPT and Claude, and human labels

Test	Result

ChatGPT, Claude, gpt-oss labelling the same 450 reviews with the human codebook	Agree with humans (kappa 0.68-0.70) as well as humans agree with each other
Same LLMs used as a "system" against the human labels	P 0.68-0.71; **InsightDesk P 0.81 is higher**, without the codebook and online
One LLM grouping a small batch at once	Groups more offline (F1 0.50) but is unstable: shuffling review order changes 46% of its groups, so it cannot keep stable cases for counting and alerts

5. Incident detection, explanations, method checks

Test	Result
Spike detection on simulated outbreaks in real streams	Original z-score detector **0%** → new detector **68-75%**, ~1 false alarm per 100 windows
Outage of vague complaints ("app not working")	**0% → 100%** detected with the unassigned-complaint alarm, confirmed on a second dataset
Cause explanations (prompt v3), two blind LLM raters	Preferred over earlier version 12:0 and 17:2; unsupported claims fell (12 → 9, 6 → 2)
Statistical method	Error-bar method validated by simulation (92-99% coverage); every result has 95% intervals
Software testing	Unit, equivalence and end-to-end tests (68% coverage, 82-100% on core modules); Postman API suite 60/60; 3 defects found and fixed

6. Questions reviewers are likely to ask (and our answer)

- **Why not just use ChatGPT/Claude?** They group well on small offline batches but change groups with input order (only 54% of groups survive a reshuffle) and cannot hold a company's full history; counting and alerts need stable cases. As labellers their precision (0.68-0.71) is below ours (0.81).
- **Recall is low on reviews (0.33).** Short reviews (21 words) look no more alike than unrelated ones; even people agree only at kappa 0.57-0.74. Re-joining recovers +26%; alert groups and the outage alarm keep incident detection high despite split cases.
- **Labels made by the team?** Benchmarks use independent Bugzilla labels (P 0.87-0.96); Play Store labels have documented agreement and a blind relabel round.
- **Similar to Oracle 2026 (retrieve-then-verify)?** Cited openly; we add online cases, re-joining, per-case incident signals and the first labelled evaluation of this design.

Still to finish before submission: end-to-end acceptance test (40 incidents), full Endava run, a "no hard gate" variant being confirmed (Firefox: recall 0.55 → 0.65, P 0.93), latency/cost per complaint, human spot check of the unseen-app labels. **Feedback welcome on:** framing, which results to lead with, target venue.